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Study Finds Fever May Lead To Improved Behavior In Children With Autism Spectrum Disorders

DECEMBER 3, 2007

Kennedy Krieger Institute Research Confirms Parent and Physician Reports that Children with Autism Spectrum Disorders Exhibit Fewer Autistic-like Behaviors during Illness with Fever

[View video of ABC 7 News story profiling research into the "fever effect."](#) ([./video/wjla_fever.wmv](#)), - January 30, 2008

[View video of WJZ 13 Health Watch story to learn more about Kennedy Krieger's latest autism research into the "fever effect."](#) ([./video/wjz_fever.wmv](#)), - January 22, 2008

(Baltimore, MD) - Over the past few decades, parents and clinicians have observed that the behaviors of children with autism spectrum disorders (ASD) tend to improve, sometimes rather dramatically, during a fever. Longer concentration spans, increased language production, improved eye contact and better overall relations with adults and peers have all been reported. In a study published today in the journal *Pediatrics*, researchers from the Kennedy Krieger Institute in Baltimore, Maryland confirmed, for the first time, parent and clinician reports that the behavior of children with ASD improves with fever. The study evaluated children with ASD during and after an episode of fever and found that fewer autistic-like behaviors were recorded for children with fever compared to controls. Understanding how fever affects the behaviors of children with ASD may provide insight into the causes of the disorder and potential treatment opportunities.

In typically developing children, signals are constantly being sent through pathways that connect the different regions of the brain and allow them to communicate with one another. Research has shown that these connections between brain regions are not made in children with autism, which limits their ability to communicate and socialize. But, the rapid behavioral changes observed with the onset of fever in children with ASD suggest that the different regions of the brain are in fact capable of connecting and communicating with one another, and that something about the fever state triggers or speeds up the signaling between brain regions. Understanding this "fever effect," including why and how connections are made between brain regions during a febrile (fever) state and not in an afebrile (without fever) state in children with ASD may provide valuable insight into the neurological basis of the disorder.

"Since autism spectrum disorders are behaviorally defined and diagnosed, studying changes in behavior resulting from a wide range of physiological changes is critical to increasing our understanding of this complex group of disorders," said [Andrew Zimmerman](#) ([/web/20150208102015/https://www.kennedykrieger.org/node/1248](http://web/20150208102015/https://www.kennedykrieger.org/node/1248)), a pediatric neurologist at the Kennedy Krieger Institute. "The results of this study are important because they show us that the autistic brain is plastic, or capable of altering current connections and forming new ones in response to different experiences or conditions."

Researchers evaluated 30 children with ASD, ages two to 18 years, during and after an episode of fever (fever was defined as 100.4 degrees F/38.0 degrees C or greater). Parents were asked to observe their child's actions and complete a standardized behavior questionnaire at three different points: during fever; when the fever subsided and the child was asymptomatic; and when the child was fever-free for seven days. These data were compared to data collected from parents of 30 afebrile children with ASD who made up the control group. Children in the control group were matched to children in the fever group in terms of age, sex and language skills. Results revealed fewer autistic-like behaviors for children with fever compared to controls, with more than 80 percent of fever subjects showing some behavioral improvements and approximately 30 percent exhibiting dramatic improvements.

"Pilot research studies such as this provide clues about the underlying metabolic changes in the brain that may prove to be targets for novel autism therapies," said Dr.

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Gary Goldstein, President and CEO of Kennedy Krieger Institute. "These and other similar findings are shaping the future direction of autism research."

Further research involving a larger participant pool is needed to better understand fever-specific effects in autism. In the future, Dr. Zimmerman would like to collaborate with other research institutions to study the underlying biologic mechanisms of fever-specific effects in autism by conducting blood tests during and after fever and analyzing immune measurements and hormonal changes in the blood. This research study was the work of senior author Dr. Andrew Zimmerman of the Kennedy Krieger Institute and the doctoral thesis of Laura Curran at Johns Hopkins Bloomberg School of Public Health and the Kennedy Krieger Institute. The study was funded by Cure Autism Now, which merged with the organization Autism Speaks in 2007.

Pediatrics is an official peer-reviewed journal of the American Academy of Pediatrics, publishing original research, clinical observations, and special feature articles in the field of pediatrics. The journal has been continuously published by the American Academy of Pediatrics since January 1948.

ABOUT AUTISM

Autism spectrum disorders (ASD) is the nation's fastest growing developmental disorder, with current incidence rates estimated at 1 in 150 children. This year more children will be diagnosed with autism than AIDS, diabetes and cancer combined, yet profound gaps remain in our understanding of both the causes and cures of the disorder. Continued research and education about developmental disruptions in individuals with ASD is crucial, as early detection and intervention can lead to improved outcomes in individuals with ASD.

ABOUT THE KENNEDY KRIEGER INSTITUTE

Internationally recognized for improving the lives of children and adolescents with disorders and injuries of the brain and spinal cord, the Kennedy Krieger Institute in Baltimore, MD serves more than 13,000 individuals each year through inpatient and outpatient clinics, home and community services and school-based programs. Kennedy Krieger provides a wide range of services for children with developmental concerns mild to severe, and is home to a team of investigators who are contributing to the understanding of how disorders develop while pioneering new interventions and earlier diagnosis. For more information on Kennedy Krieger Institute, visit www.kennedykrieger.org.

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